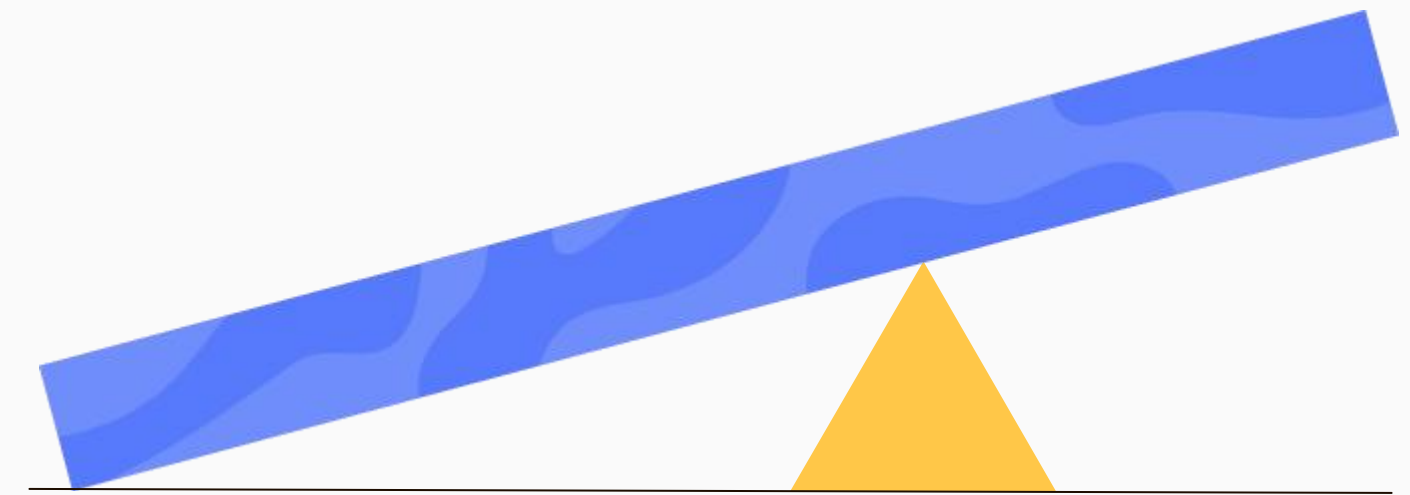


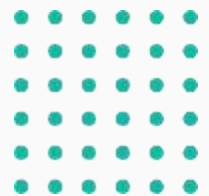
Apex R.R.S

by Apex Neural System





01 Problem Statement



What is the problem?



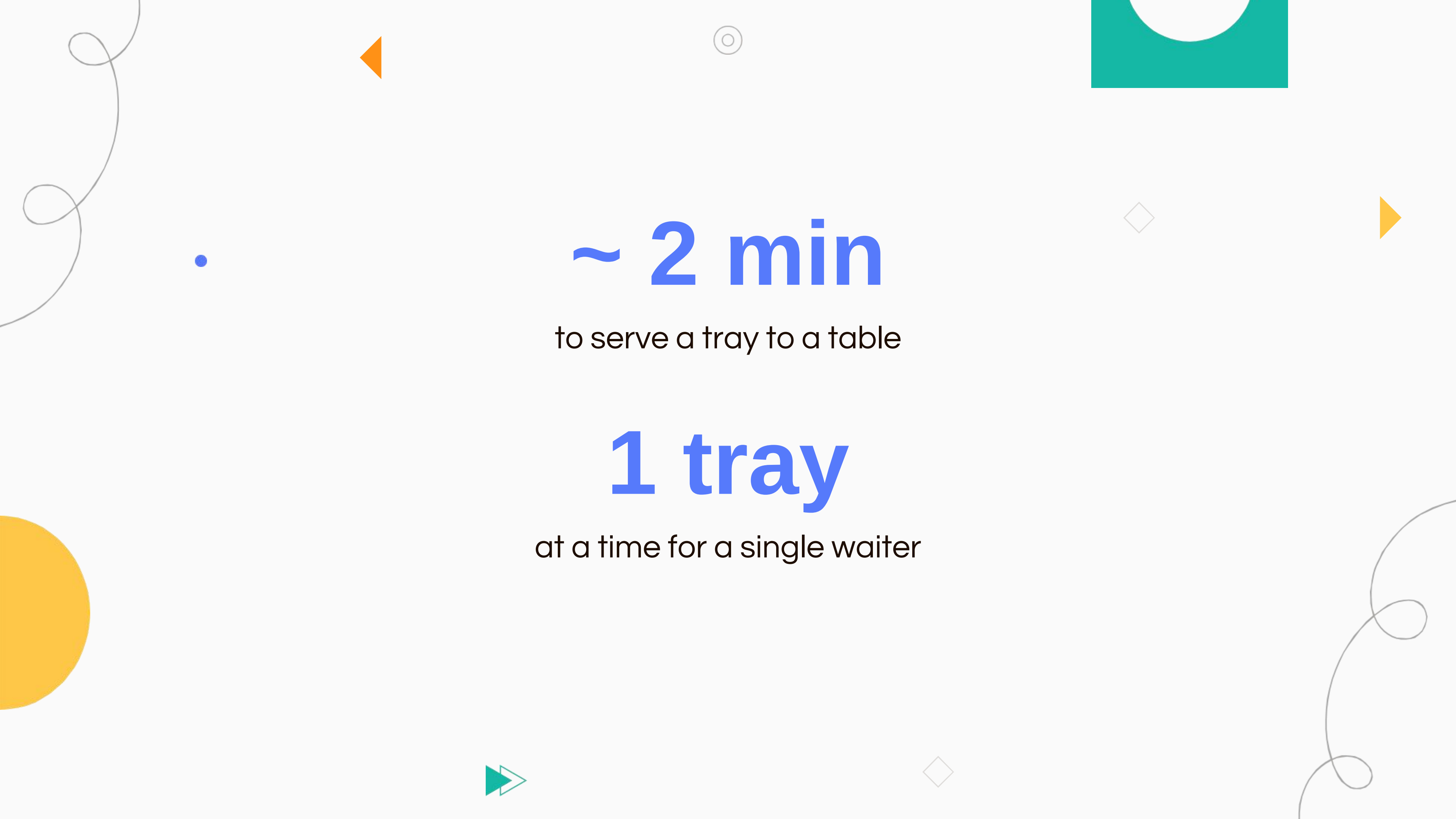
Slow Food Delivery

Lack of manpower during peak
hours



Labor shortages

Required labor



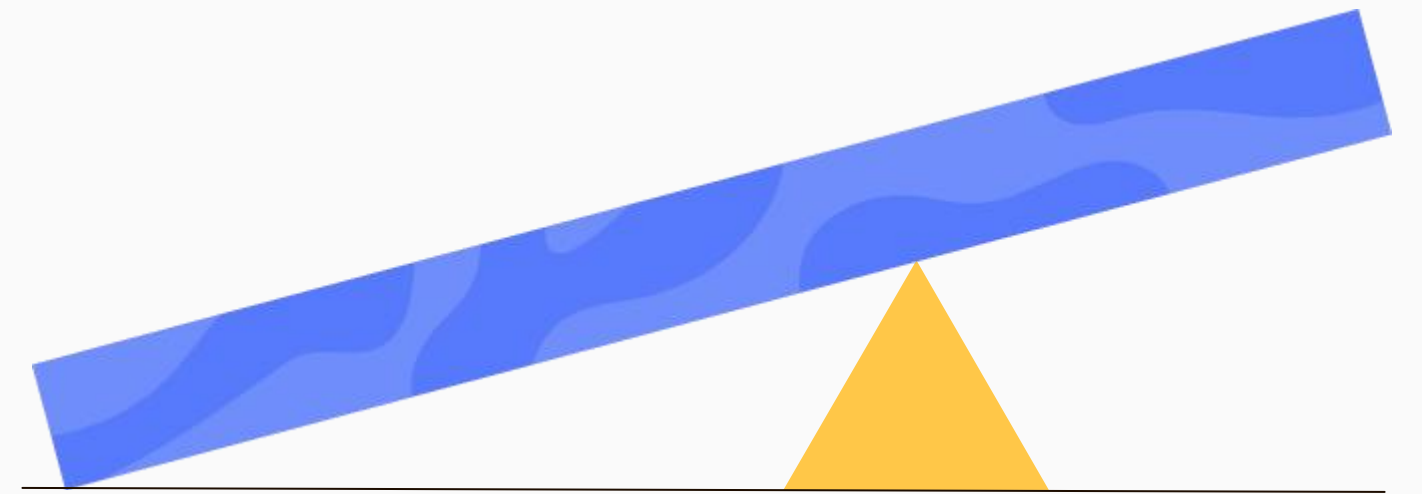
~ 2 min

to serve a tray to a table

1 tray

at a time for a single waiter

02 Proposed Solution



How it works (Steps)

Chef will select the which
table for the robot waiter
to serve

Step 1

The robot hand will grab
the food and place it onto
the table

Step 3

Step 2

The robot waiter will be
send to that table

Step 4

The robot waiter move
back to the starting point





~66%

of manpower reduced

up to 3 tray

can be carried by Neural RRS instead of 1
tray by a waiter

Why Our Solution?



Reduce Manpower

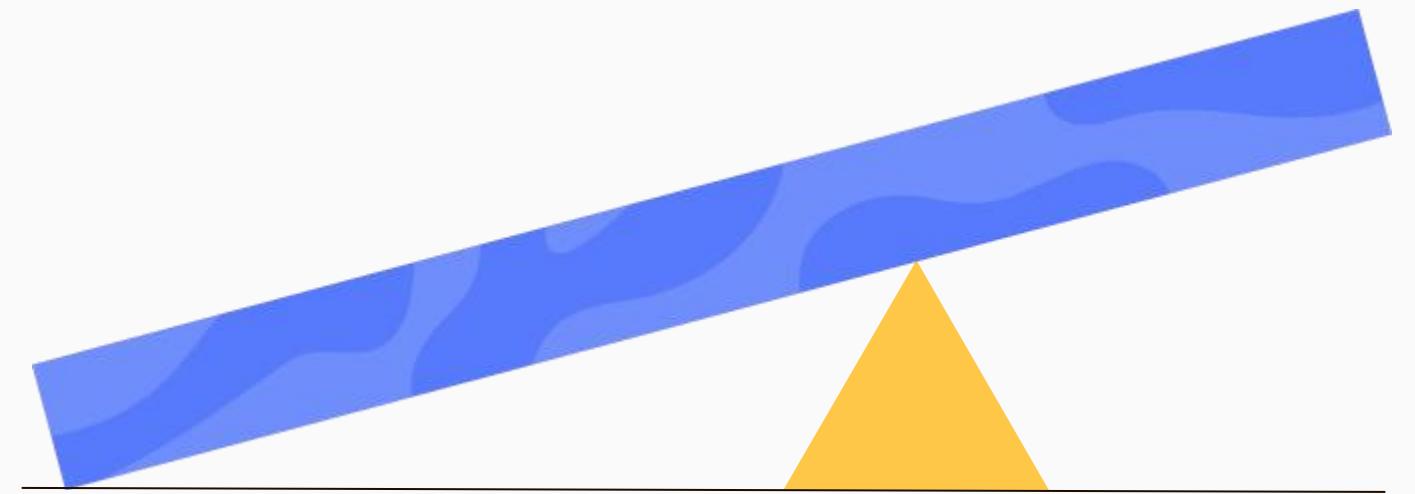
A robotic car for food delivering and robotics arm for handling grabbing food.

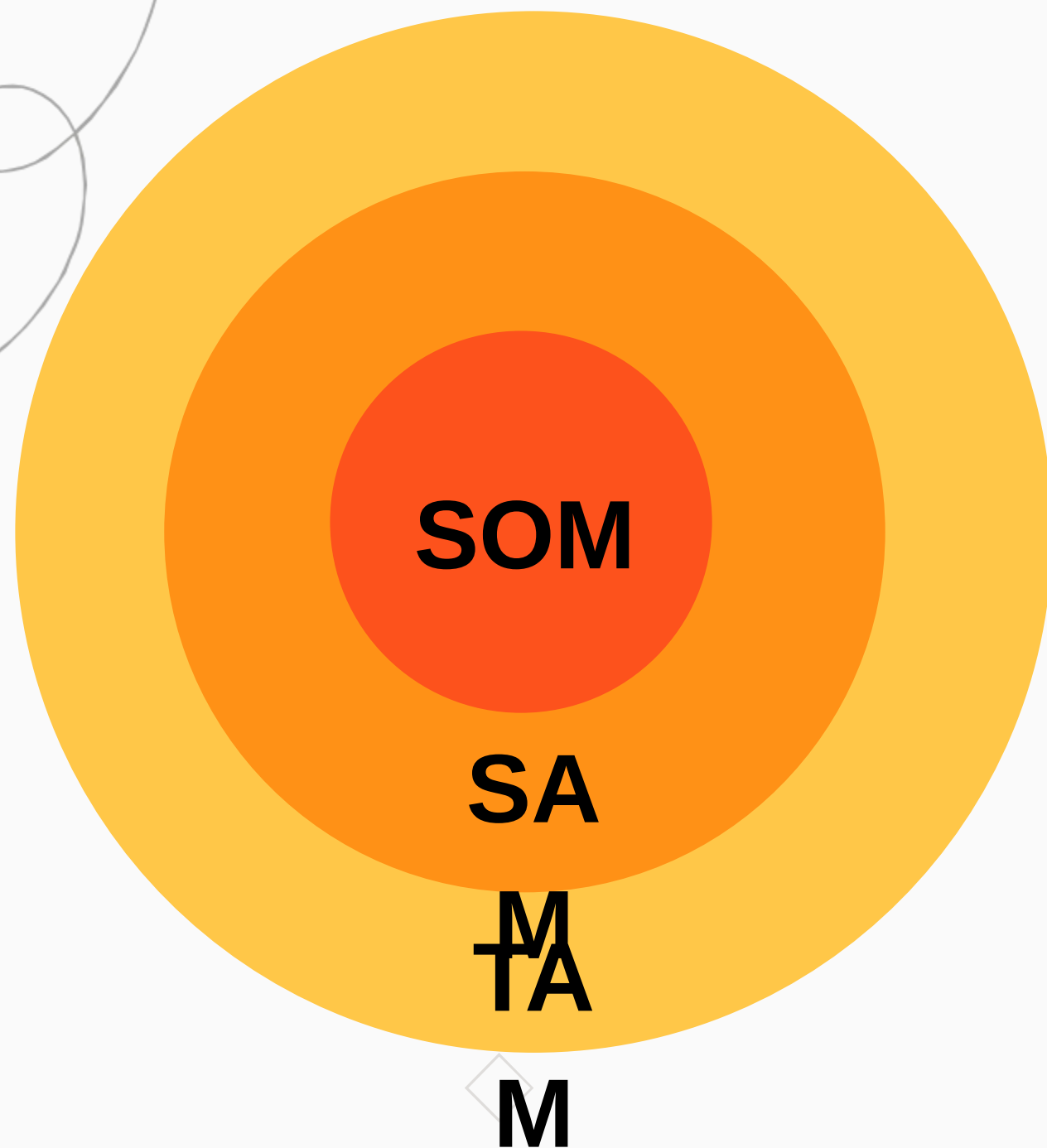


Suitable for Repetitive Task

Designed to reduce the labour needed for serving food repetetatively

03 Market Size





TAM - \$4.2B

Total global restaurant service robotics market

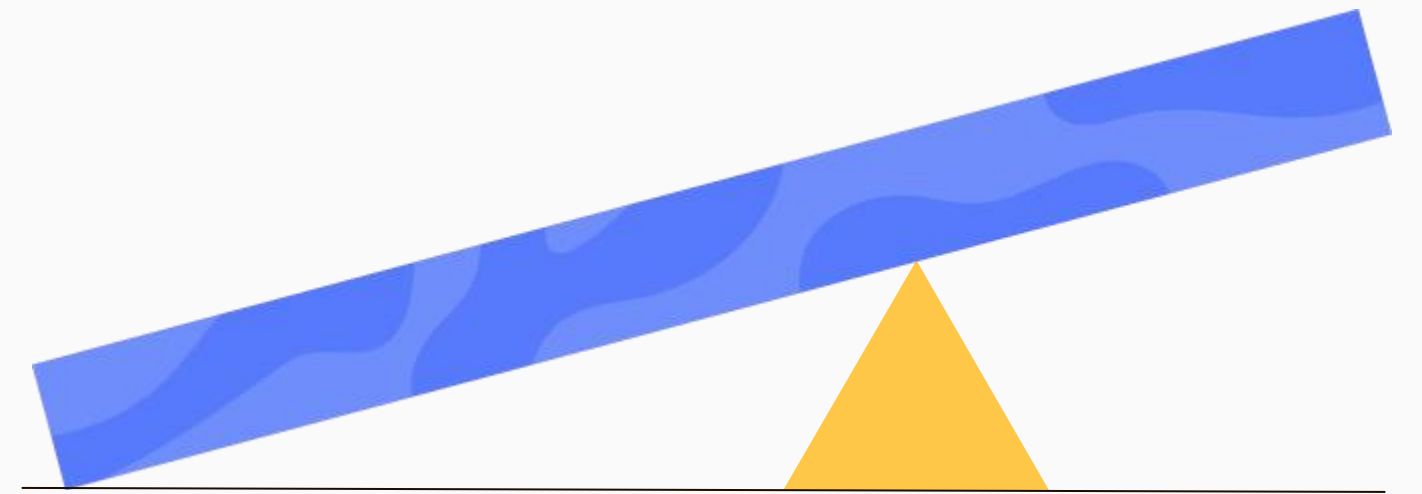
SAM - \$1.6B

Total Asia Pacific restaurant service robotics market

SOM -

\$16M
Capture 1% of the restaurant robotics market within Asia Pacific.

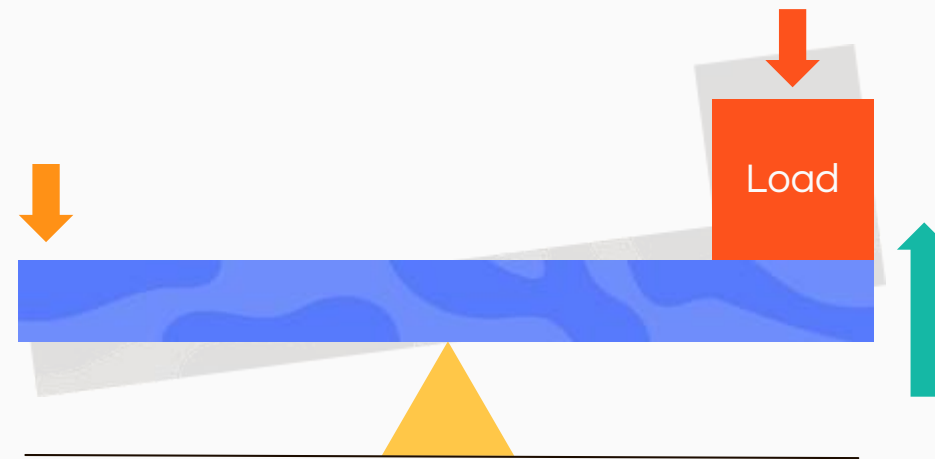
04 Business Model



Business Model

Robotics-as-a-Service (RaaS)

Monthly fees per robot
RM2000/unit for 2 years



Onboarding & Setup

RM2000 once for integration
fees covering POS/ kitchen
system setup & floor mapping

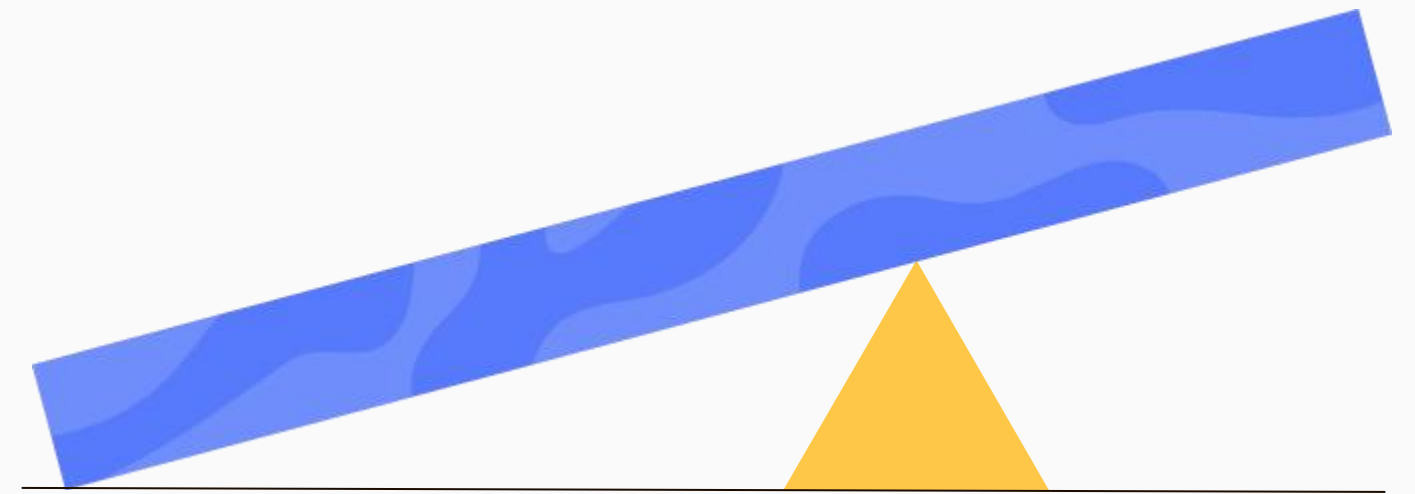
Estimation of ROI

Item	Cost (RM)
Labour Cost (RM2500 * 2 person * 12 months)	60,000
Apex RRS Subscription (RM2000 * 2 unit * 12 months)	48,000
Apex RRS Setup Fees	2,000
Cost Saved	10,000

$$\begin{aligned}\text{Annual ROI} &= \text{RM}(10000 / 50000) \times 100 \\ &= 20\%\end{aligned}$$

05

Road Map



Roadmap

Deliver robot and live testing on
few restaurant

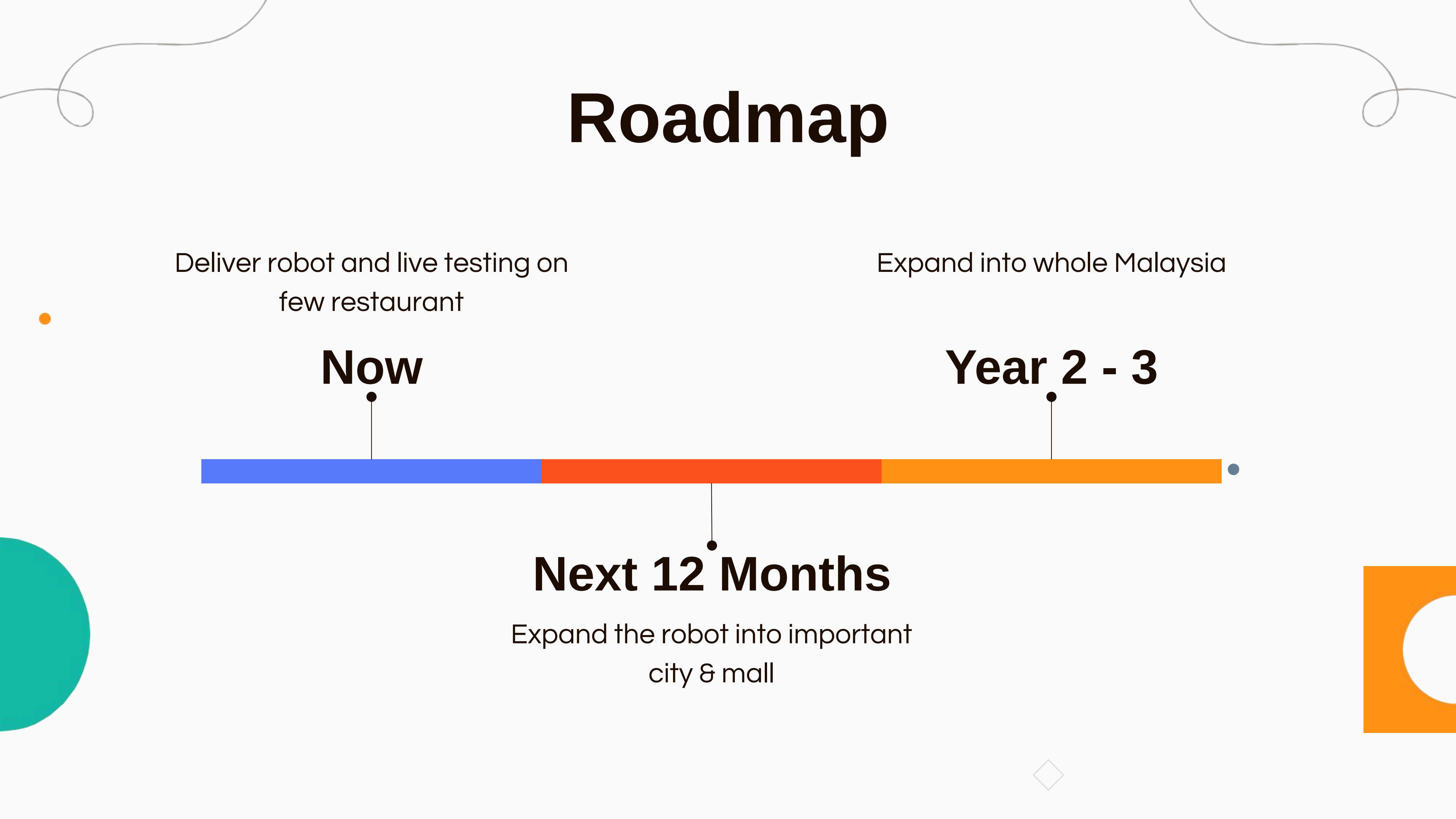
Now

Expand into whole Malaysia

Year 2 - 3

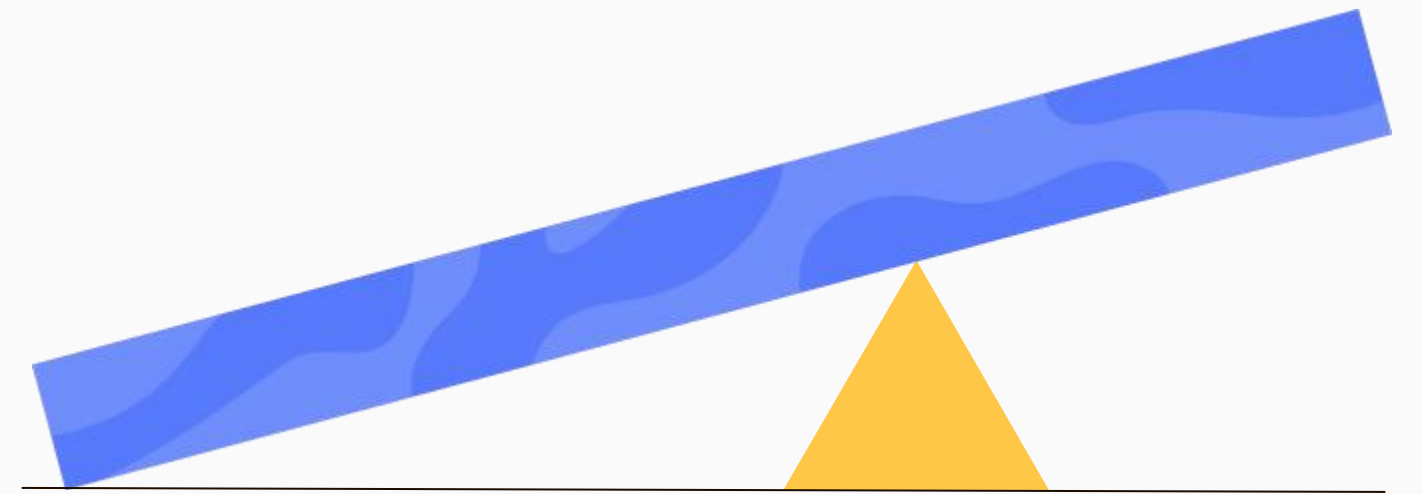
Next 12 Months

Expand the robot into important
city & mall



06

Technical Part





Objective

Technical Objective & Success Criteria



Technical Objective



To **safely** and **autonomously** deliver food to the **correct** customer table using a **mobile robot** and **robotic arm**.

Success Criteria



Correct food reaches the correct destination



Mobile robot navigates safely and avoids obstacles



Robot arm starts automatically at the docking point



Food is transferred safely and securely



System returns to standby and ready for next task

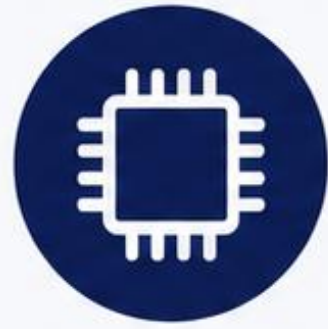
Complete System Architecture

How the robot thinks and responds



SENSE

- Detect food tray / order tag
- Detect people and obstacles
- Confirm arrival at customer table



DECIDE

- Identify the correct table
- Choose a safe route
- Decide when to trigger the robot arm



ACT

- Deliver food to the table
- Stop at the docking position
- Robot arm transfers the tray



VERIFY

- Confirm correct table reached
- Confirm tray placed safely
- Return to kitchen / standby

Sensors detect → controller decides → robots serve → system checks again

System Workflow

How our restaurant robot works



Robot Arm Features & Supporting Technologies

Automatic Tray Transfer for Restaurant Service



Product Features



Automatic Tray Pickup



Precise Tray Placement



Stable & Safe Gripping



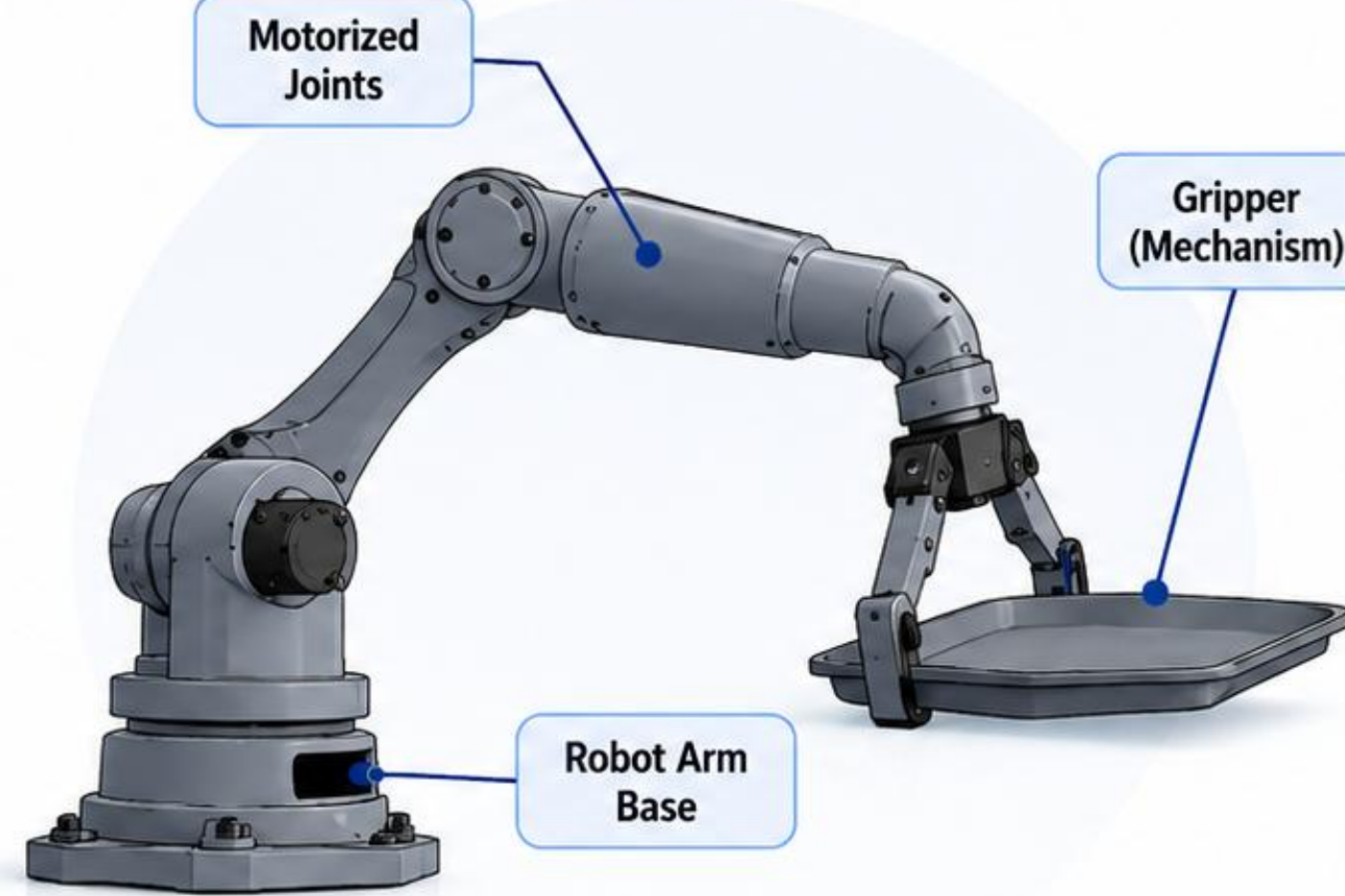
Repeatable Motion



Table-Side Serving Support



Safe Food Transfer



How it transfers the tray

1



Robot arrives at docking point

2



Robot arm automatically grips the tray

3



Moves and places tray on customer table

4



Returns to standby position



Supporting Technologies



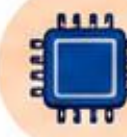
Servo / Actuator Motors



Gripper Mechanism



Multi-Joint Arm Structure



Controller Board



Docking / Proximity Sensor



Position Control & Safety Limit



Our Key Difference



Industrial delivery robots: transport trays for manual unloading



Our concept: transport + automatic robotic-arm tray transfer

Product Features & Supporting Sensors

Functions on one side, sensors on the other side

★ Product Features



1 Autonomous Navigation



2 Obstacle Avoidance



3 Food Delivery



4 Tray Transfer



5 Stable Movement



6 User-Friendly Operation



Supporting Sensors / Technologies



1 LiDAR Sensor



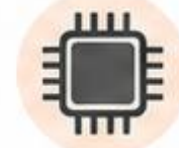
2 Front Obstacle Sensor



3 Tray Presence Sensor



4 Drive Motors



5 Main Controller



6 Battery System

Sensor Architecture Defense

Why each sensor was selected — and what it cannot do



ROBOT SENSOR LOCATIONS



One sensor cannot handle every situation.
Each one covers a different risk.



LiDAR Sensor

Main navigation and mapping

REQUIREMENT

Map the restaurant and navigate safely.

CHOICE

2D LIDAR + SLAM



WHY

Wider view than ultrasonic sensors, and less affected by lighting than cameras.



TRADE-OFF

A 2D LiDAR scans at one fixed height, so it may miss very low obstacles.



Tray Presence Sensor

Checks the tray before movement

REQUIREMENT

Confirm the tray before the robot moves.

CHOICE

Photoelectric sensor



WHY

Non-contact detection causes less wear than a mechanical switch.



TRADE-OFF

Dust, spills or dark trays may affect the reading.



Front Obstacle Sensor

Extra close-range safety layer

REQUIREMENT

Detect close and low obstacles.

CHOICE

3 ultrasonic sensors



WHY

Affordable short-range safety with wider front coverage.



TRADE-OFF

Soft or angled objects may produce weak echoes.



KEY MESSAGE

LiDAR **guides** the robot, ultrasonic sensors **protect** the front, and the tray sensor **ensures safe delivery**.

Evidence: Benchmark-Informed Performance Targets

Simple measurable targets for our restaurant waiter robot prototype



Commercial Benchmark

- 400 deliveries per day
- Always punctual



Obstacle Safety

- Detection target: 50–100 cm
- Safe stopping distance: ≥ 20 cm
- Success target: 9/10 trials
- Objects: person, chair, box, low obstacle



Delivery Capacity

- Estimated cycle time: 2–2.5 min
- Practical capacity: 15–18 trays/hour
- Assumption: 20 m round trip
- One tray per cycle



System Reliability

- Planned test: 10 full sequences
- Target success: 80–90%
- Includes navigation, docking and tray transfer
- No collision or dropped tray



Use benchmark data to set realistic prototype targets, then validate with physical testing.

Current Limitation & Validation Plan

Being transparent about what still needs real-restaurant testing



Current Limitation

Fixed docking and tray alignment

The prototype relies on a fixed docking position and consistent tray placement for successful robot-arm pickup.



Impact

Pickup accuracy depends on alignment

Docking error or tray movement may reduce pickup accuracy and interrupt the full delivery sequence.

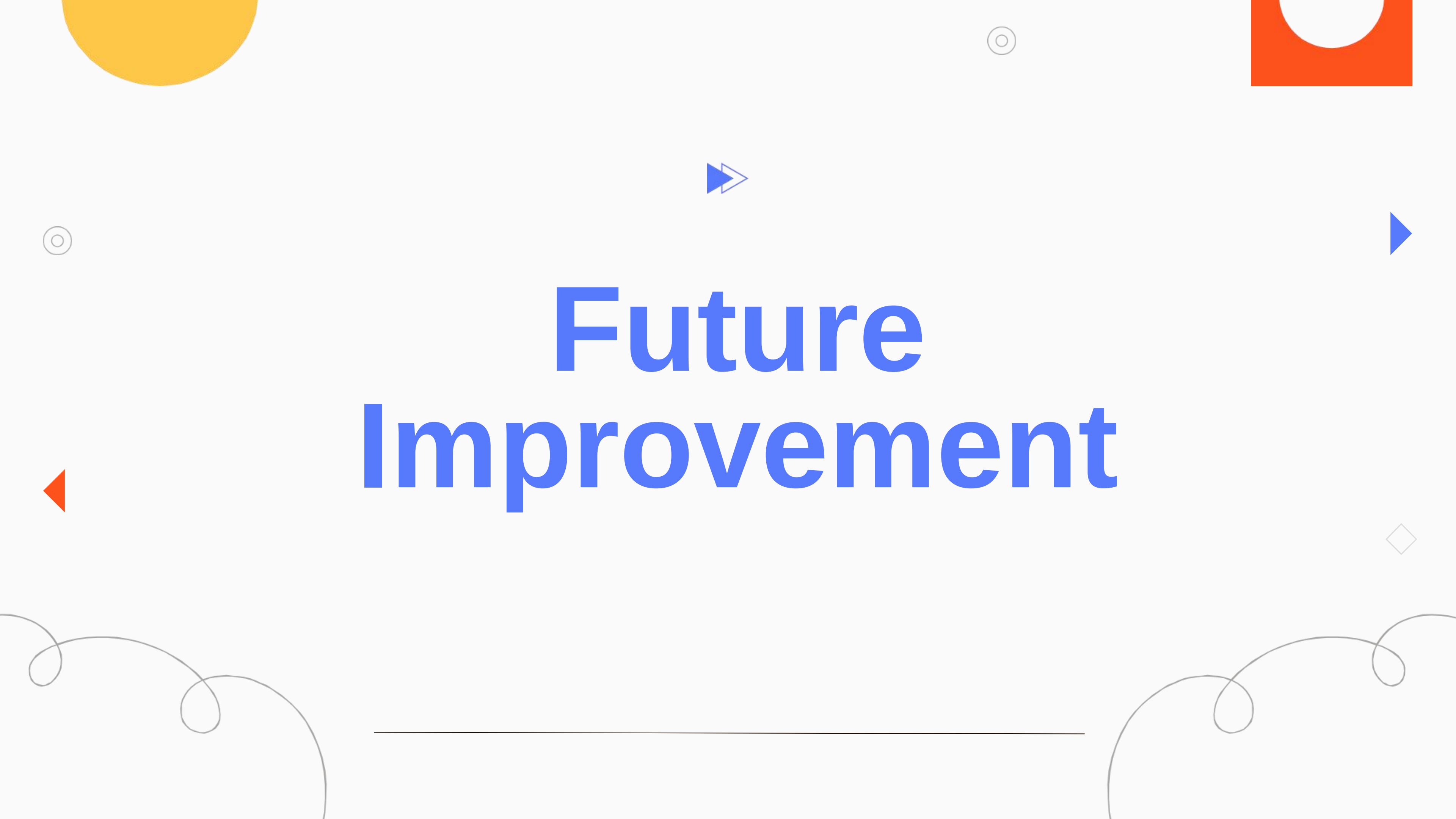


Next Validation

How we will prove the system in a real restaurant setting

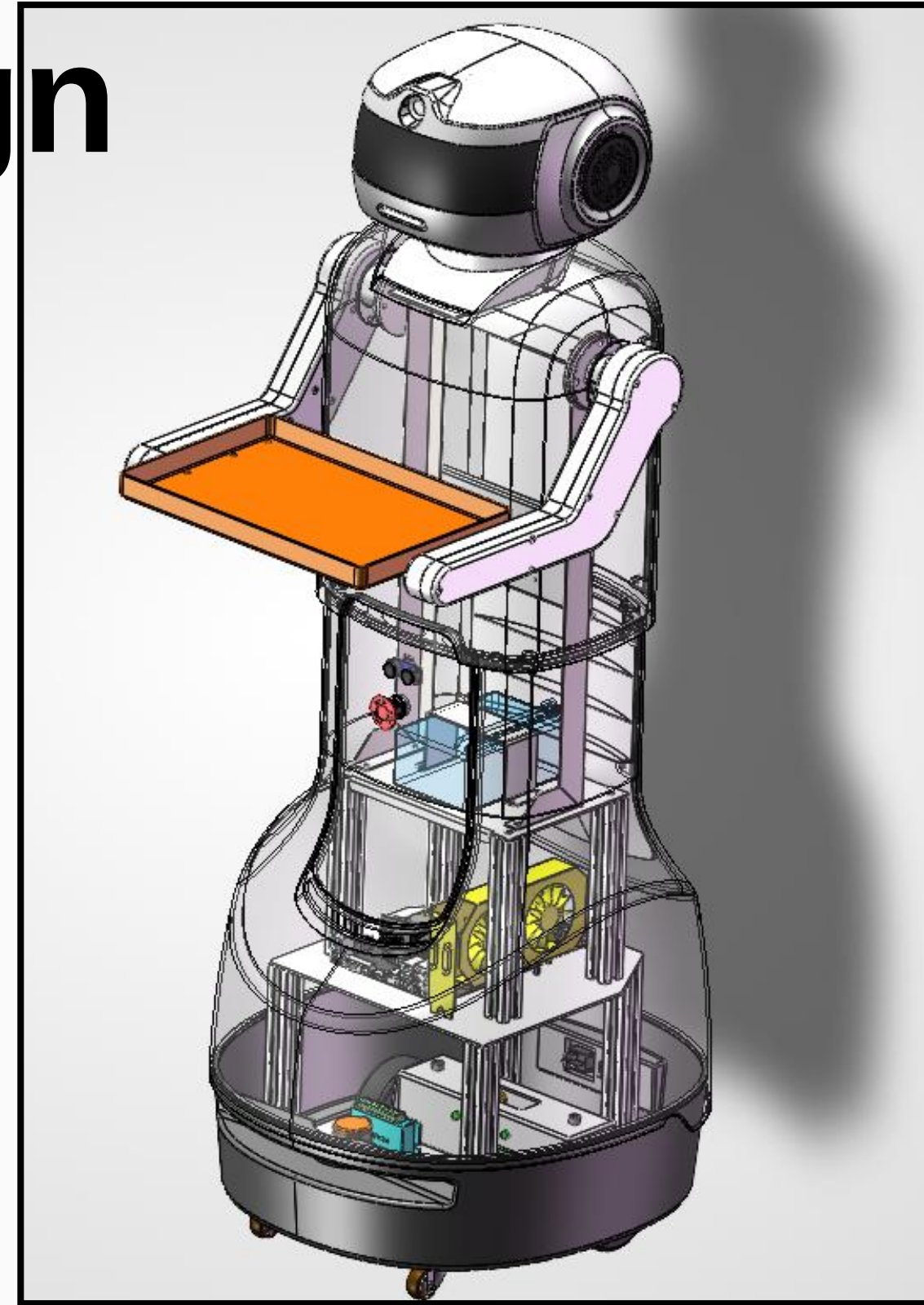
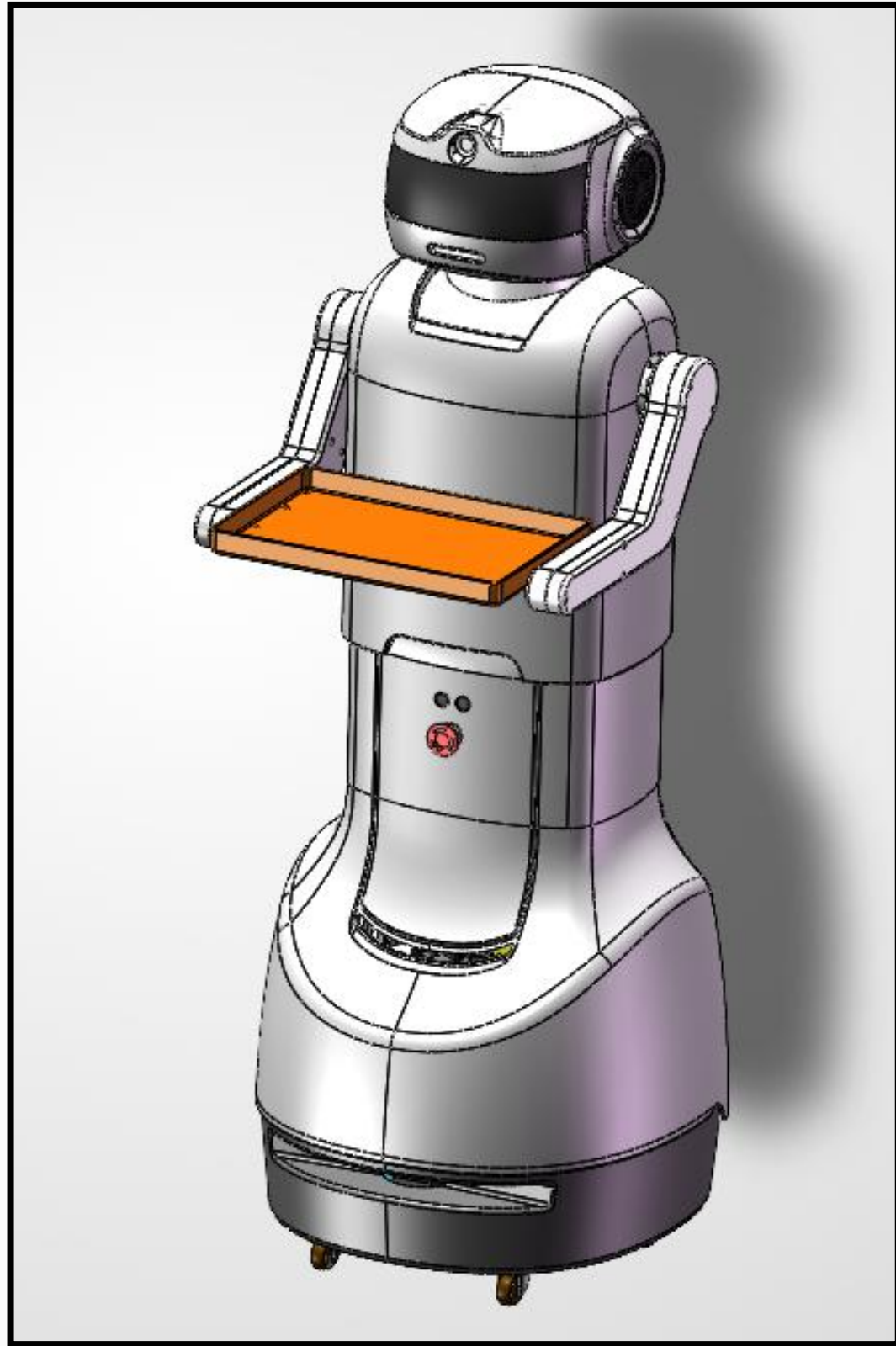
- 30 Conduct 30 complete delivery and tray-transfer trials
- cm Measure mobile robot docking error
- kg Test different tray positions and tray weights
- ↔ Test moving people and changing obstacles
- % Record pickup success rate and cycle time

The core concept is feasible under fixed conditions; real restaurant testing is required before commercial deployment.

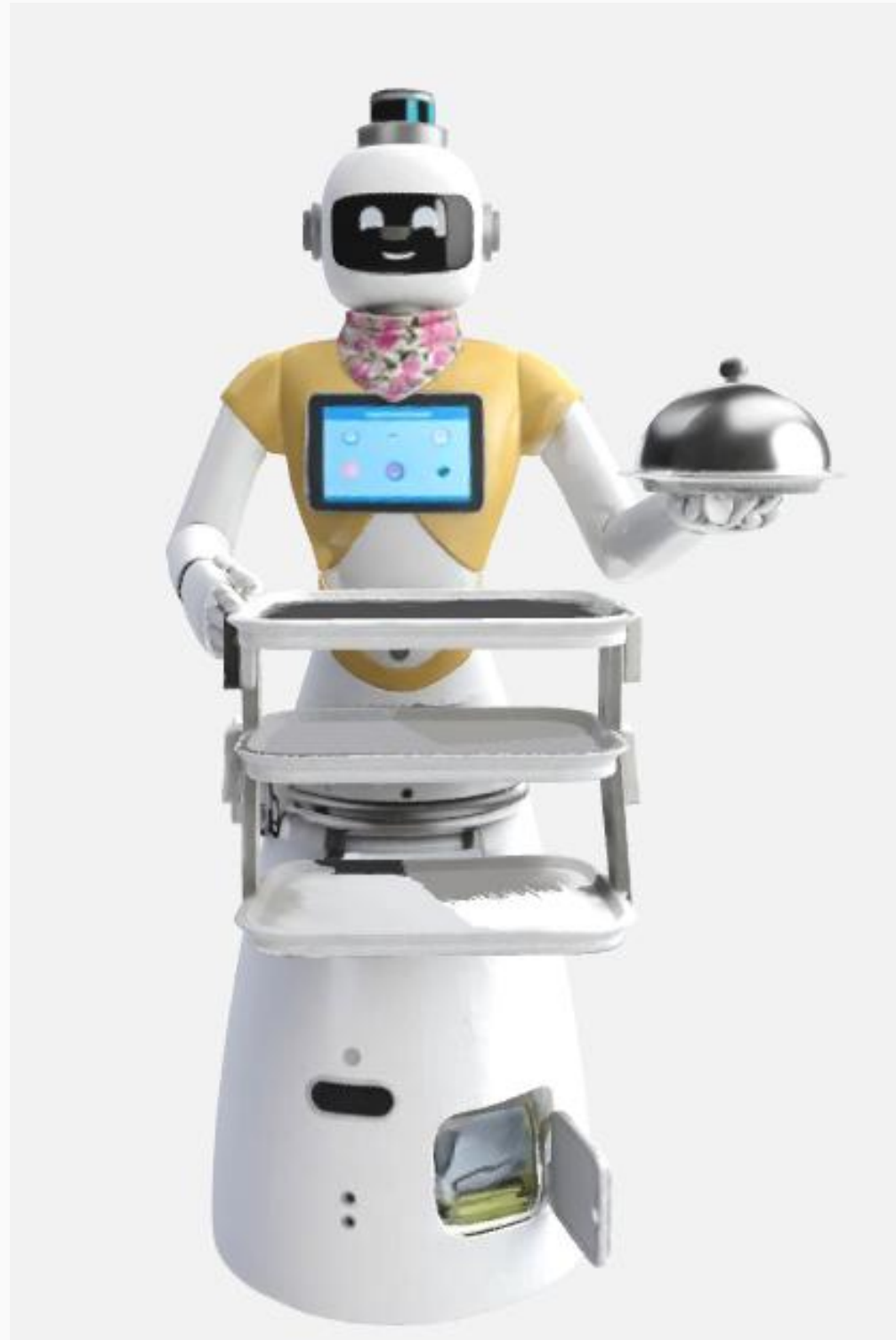


Future Improvement

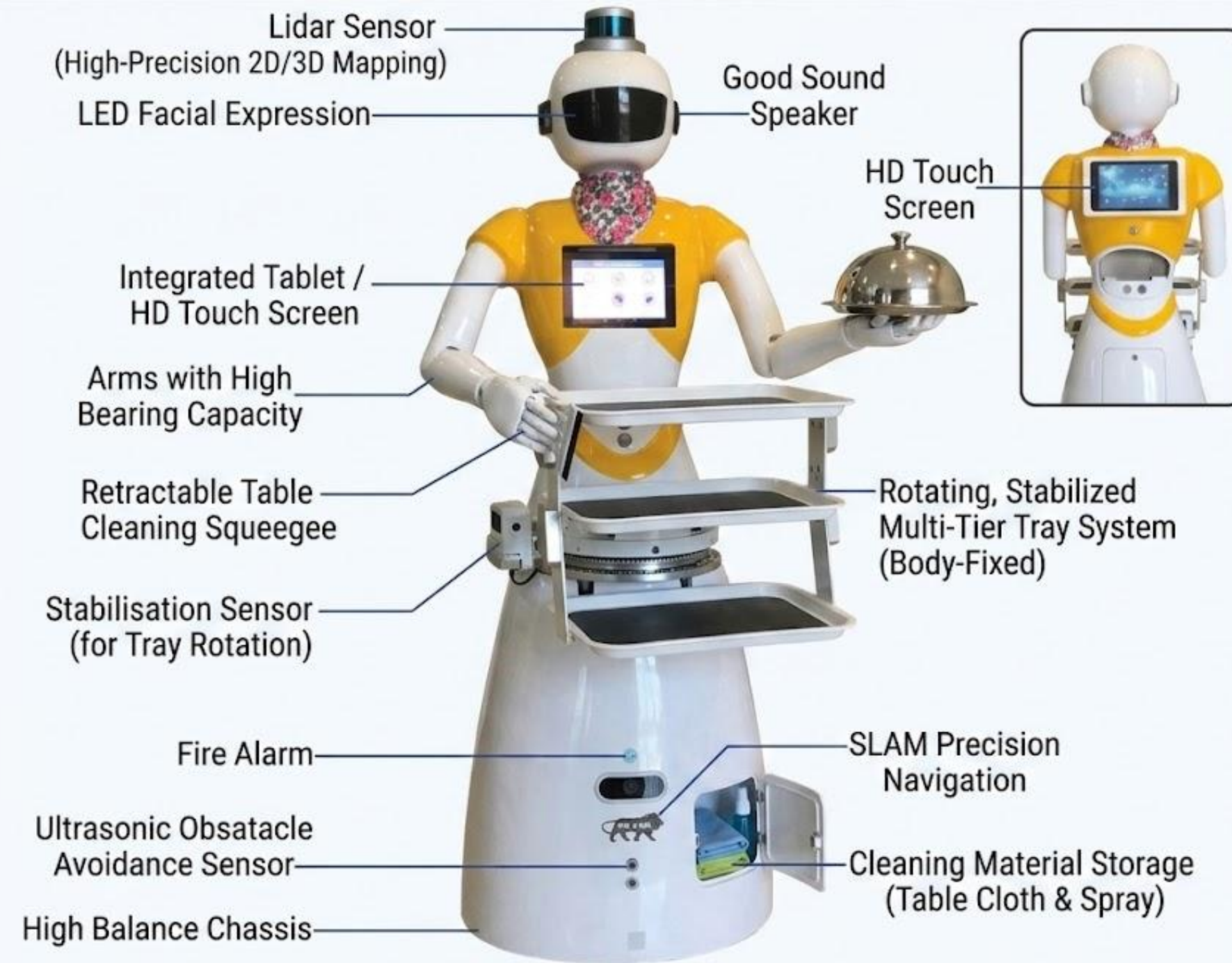
Conceptual Design

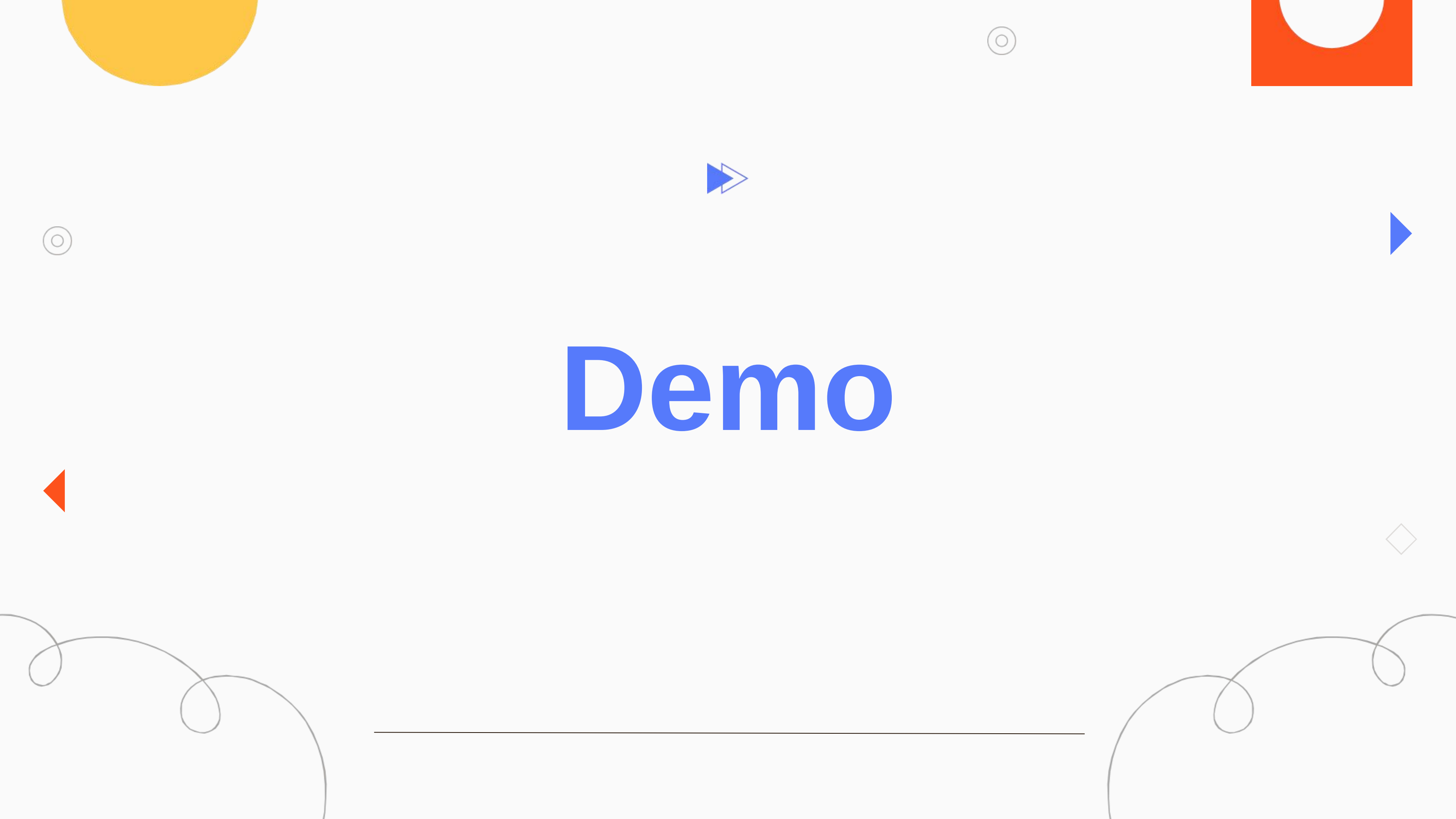


Future Improvement View

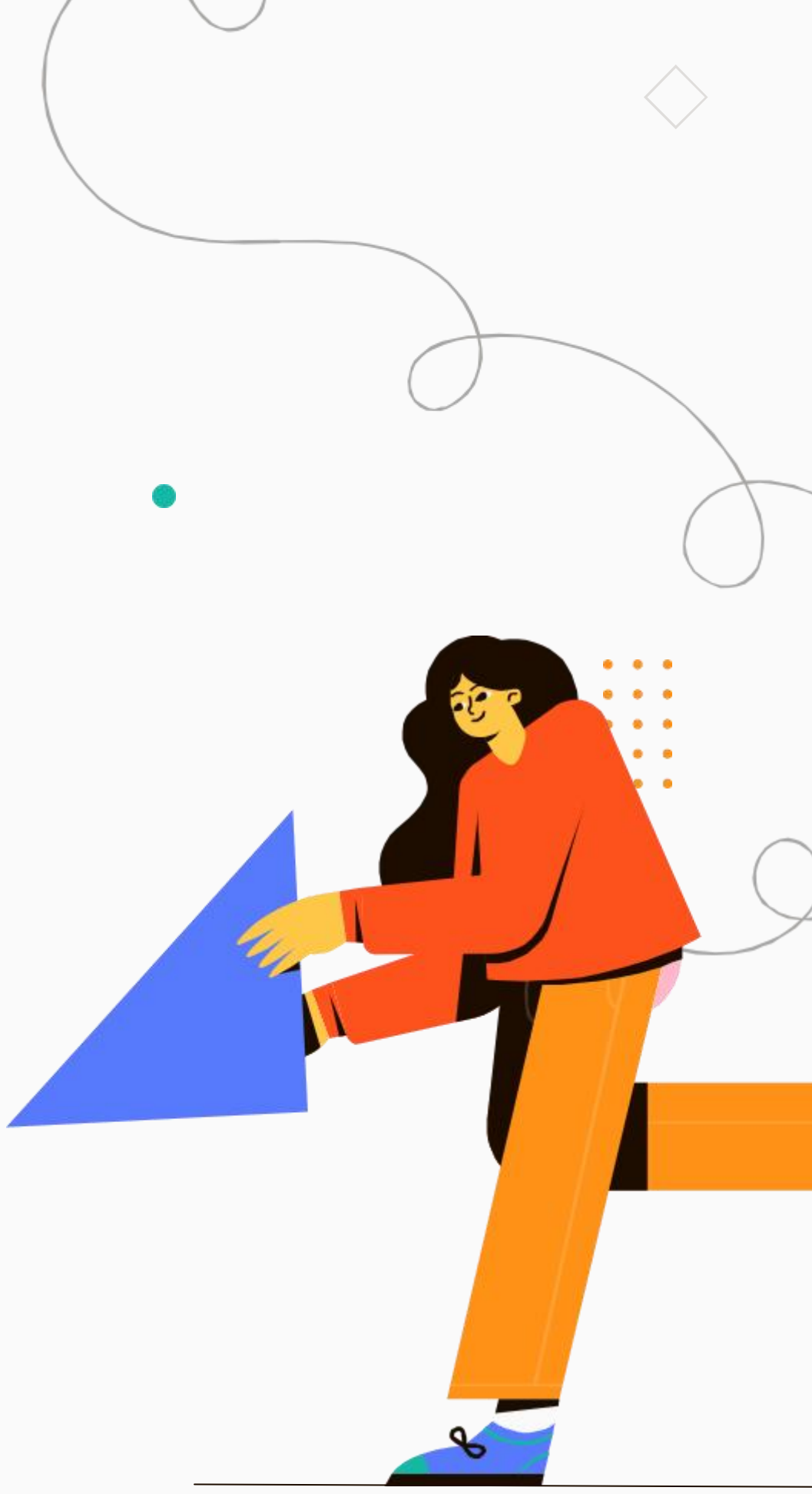


Ruby – The Robot Waitress





Demo



Thanks!

Do you have any questions?

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